

SUNBLAZER III



Empowering Off-Grid Communities



The SunBlazer III expands the generation capacity of the SunBlazer Lite to drive Microgrids and larger industrial loads.

The SunBlazer III (SB III) aims to service larger villages cost-effectively at the same time introducing a simple scaling model with added reliability through redundancy. Numbers of systems can be deployed throughout a larger community to provide many kilowatts of power as the community grows its demands. The systems will come with a standardized Microgrid design scalable to load requirements for in-country tailored design and deployment.



How it Works

SB III is a mobile solar 795W to 2,385W base station including 100-200Ah at 24V station batteries, PBKs, for home lighting kits. Each PBK has an auxiliary outlet for other DC powered devices in the home such as cellphones or small entertainment devices.

SB III can provide all the capacity of SB Lite or SB II plus the addition of a Microgrid of 1.6kW capacity. The heavy duty plastic box containing the electronics can rest under the frame, or on a concrete pad nearby. The assembly needs protection by a security fence, or alternatively the box could go indoors.

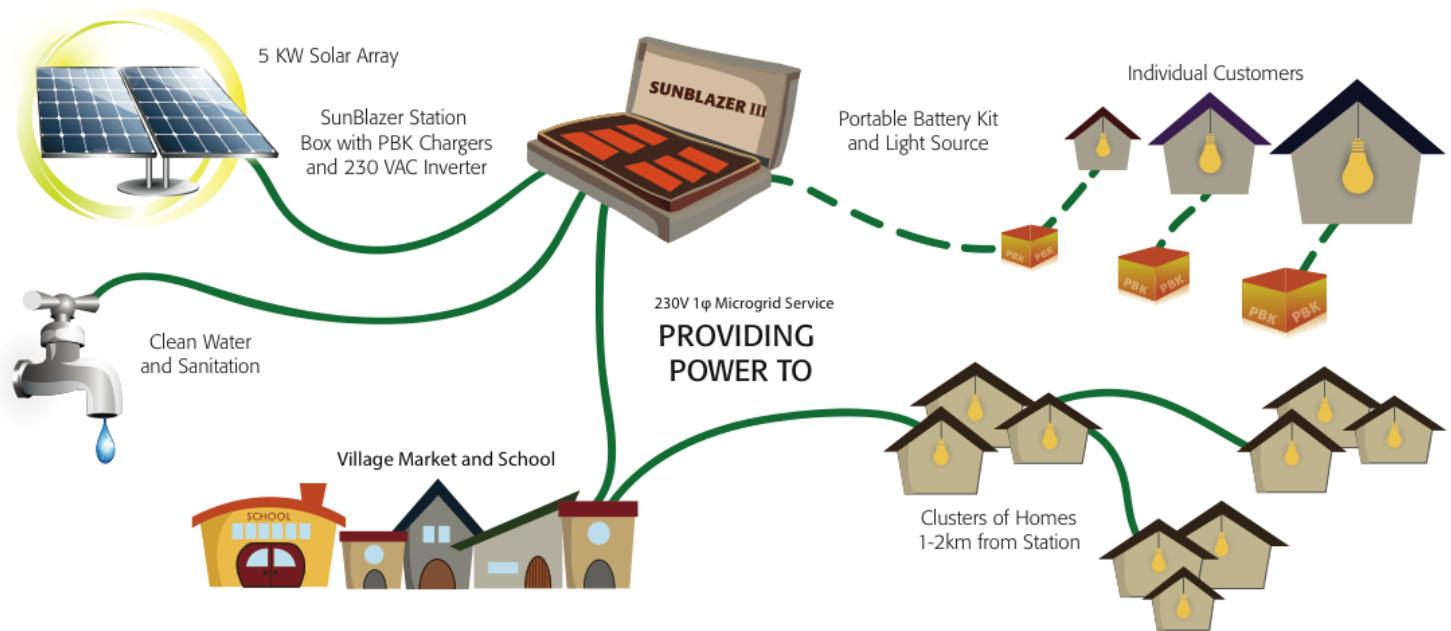
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THE SUNBLAZER III AT WORK



SUNBLAZER III UNIT TECHNICAL SPECIFICATIONS

SPECIFICATIONS	MECHANICAL
Dimensions	Transport 96"L x 48"W x 70"H Setup (3 to 9 solar panels) 83" to 197"L x 80" W x 78"H (PV panel pivot point)
Weight	950 lbs. 431 kilos - 3 solar panels, 2 station batteries
Sun Angle Adj.	-45 to +45 deg.
Terrain Requirement	Flat
Frame Construction	Legs-Powder coated steel, Header-Galvanized
Enclosure Construction	High-density polyethylene (HDPE) plastic, powder-coated steel hinges. One per 3 solar panels
Mobility	Self-contained shipping crate
Security	Electronics and station batteries are accessible only by operator
Typical Setup Time	1 hour with 2 people (3 panels)

SPECIFICATIONS

ELECTRICAL

Solar Panels	Base Unit: 3-265 watt panels for 795 watts total Expanded Unit: 9-265W panels for 2,385W total
Solar Generation (3, 6 or 9 panels)	Base Unit: 3.6 kWh/day in equatorial regions Expanded Unit w/9 panels: 10.7 kWh/day
Station Batteries	Base Unit: 2-12V 100Ah FLA flooded lead acid, 24 volt bus Expanded Unit: 2-12V 200Ah FLA, 24V bus. <u>One base unit for every 3 panels</u>
Solar Charge Controller	3 stage PWM, temperature compensated, automatic load disconnect if batteries are <50% charged and re-connection upon charge
Sub Chargers for Universal Portable Battery Kits (PBK's)	6 charging channels for 12V 12 or 18 Ah UPBK batteries, or 10 charging channels for 10-12Ah units simultaneously. Each channel has 3 stages with LED indication of completed charge. Each channel has on-demand "restore charge" as needed to help extend cycle life
PBK Charge Time/Capacity	4-hour charge time with capacity to support 2 shifts of 6-10-12Ah or 6-18Ah UPBK's per solar day
AC Output (optional)	1000 watts at 230 VAC 50 or 60 Hz (expandable upon request)
AC Input (optional)	230 VAC battery charger if standby generator or grid AC is accessible



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